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Automatic identification of hard and soft tissue landmarks in cone-beam computed tomography via deep learning with diversity datasets: a methodological study

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Abstract

Background Manual landmark detection in cone beam computed tomography (CBCT) for evaluating craniofacial structures relies on medical expertise and is time-consuming. This study aimed to apply a new deep learning method to predict and locate soft and hard tissue craniofacial landmarks on CBCT in patients with various types of malocclusion.

Methods A total of 498 CBCT images were collected. Following the calibration procedure, two experienced clinicians identified 43 landmarks in the x-, y-, and z-coordinate planes on the CBCT images using Checkpoint Software, creating the ground truth by averaging the landmark coordinates. To evaluate the accuracy of our algorithm, we determined the mean absolute error along the x-, y-, and z-axes and calculated the mean radial error (MRE) between the reference landmark and predicted landmark, as well as the successful detection rate (SDR).

Results Each landmark prediction took approximately 4.2 s on a conventional graphics processing unit. The mean absolute error across all coordinates was 0.74 mm. The overall MRE for the 43 landmarks was 1.76 ± 1.13 mm, and the SDR was 60.16%, 91.05%, and 97.58% within 2-, 3-, and 4-mm error ranges of manual marking, respectively. The average MRE of the hard tissue landmarks (32/43) was 1.73 mm, while that for soft tissue landmarks (11/43) was 1.84 mm.

Conclusions Our proposed algorithm demonstrates a clinically acceptable level of accuracy and robustness for automatic detection of CBCT soft- and hard-tissue landmarks across all types of malformations. The potential for artificial intelligence to assist in identifying three dimensional-CT landmarks in routine clinical practice and analysing large datasets for future research is promising.

Keywords 3D landmark identification, Accuracy, CBCT, Deep learning

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Background

Cephalometric analysis is a vital aspect of the daily workflow for orthodontists and maxillofacial surgeons. Accurate and efficient identification of anatomical landmark is crucial for providing morphometric guidelines used in diagnosis, surgical planning, growth analysis, and treatment planning by analysing dental and skeletal relationships in the craniofacial complex [1].

With the advent of artificial intelligence (AI), automatic measurement of two-dimensional (2D) lateral cephalograms has emerged, an alternative to manual positioning, which is both time-consuming and dependent on the expertise of medical professionals. Automated landmark localisation and cephalometry analysis offers enhanced stability and repeatability and have already demonstrated superior average prediction accuracy [2–4]. However, due to their limited dimensions, overlapping of bilateral tissues, unclear anatomical structures, and varying magnifications resulting in different proportions of certain structures, 2D cephalometric lateral films inevitably have some degree of error and a lack of accuracy in anatomical identification and diagnostic comprehensiveness [3]. Since its introduction in the 1990s, cone-beam computed tomography (CBCT) has become a widely used clinical practice due to its low radiation dose, high resolution, and ability to provide multidimensional information on craniofacial hard tissues. CBCT can directly measure the widths of the upper and lower jaws without being affected by tooth tilt or displacement. Consequently, it is regarded as the gold standard for diagnosing maxillary width discrepancies [5]. Additionally, CBCT is regarded as the preferred non-invasive method for nasopharyngeal airway assessment [6] and can more accurately determine the degree of jaw deviation without soft tissue interference. Therefore, CBCT allows clinicians to assess complex maxillomandibular deformities and craniofacial anomalies, thereby improving diagnosis and treatment planning when the benefits of the diagnosis and/or treatment plan outweigh the potential risks of exposure to radiation [7–11]. Despite the importance of three-dimensional (3D) CT imaging, manual identification of anatomical landmarks in images of highly complex 3D structures, such as the skulls, remains a challenging and laborious task. It is prone to random and systematic errors, leading to inconsistencies within and between assessors. Therefore, developing an automated and comprehensive 3D-CT landmark localisation system is essential.

In recent years, several studies have reported the development of 3D-AI auto-localisation techniques, with the accuracy of automatic landmark identification consistently improving due to iterative updates to the algorithms [12]. In conventional 2D measurements, a distance of 2 mm or less between automated and manual recognition is considered clinically accurate; 4 mm is deemed acceptable. However, the 3D Euclidean distances (EDs) are derived from three coordinate directions, and an established standard for the accuracy of 3D automated landmark identification is currently lacking [13]. Furthermore, the studies employed a limited number of 3D-CT datasets, with limited sample types and a restricted number of landmarks (primarily hard tissue anatomical structure points, predominantly vertical and sagittal diagnostic points), resulting in a restricted scope of diagnostic information [12, 14–18]. Consequently, the comprehensiveness, universality, accuracy, and robustness of the algorithms remain to be upgraded [12, 14, 16, 17, 19, 20]. Therefore, it is crucial to develop a more precise, comprehensive, rapid, and suitable method for 3D diagnosis of various types of malocclusion through automatic landmark identification.

It is well established that the quantity and quality of the dataset are key determinants of the accuracy and robustness of the algorithms used. The study aimed to enhance the precision and generalisation of automatic landmark identification in 3D-CT by increasing the number of datasets, expanding the scope and diversity of cases (including different dentition types and patients with cleft lip and palate, syndromes, and condylar fractures), incorporating additional landmark categories (especially for diagnosing transverse maxillary discrepancies and soft tissue key points), and utilising the latest attention model optimisation algorithms. It is anticipated that AI will eventually enable the automatic identification of 3D imaging data for diagnosing dental and maxillofacial deformities.

Methods

Review and comparison of relevant literature

In order to source studies that are relevant to the present study for better design and comparison of the results, we performed a search in December 2023 without time restrictions that combined MeSH terms and free text words using Boolean operators ('AND', 'OR') on these databases: PubMed, Scopus, Web of Science, and Cochrane Library (for Cochrane reviews). We used the following search terms to find the articles: 'automatic*' AND 'cephal*' AND ('3D' OR 'CBCT' OR 'CT') AND ('artificial intelligence' OR 'machine learning' OR 'deep learning' OR 'learning'). Additionally, we performed a thorough manual search of references within original articles, reviews, and conference proceedings to identify any additional studies that were not retrieved from the chosen electronic databases.

Studies were selected according to the criteria outlined below: (1) Study designs: we included in vitro and in vivo prospective and retrospective studies (clinical trials, comparative studies, validation studies, or evaluation studies). We excluded book chapters, animal studies, case reports, epidemiologic studies, narrative reviews, and author opinion articles. (2) Subject of the study (automatic landmark identification for 3D CBCT or CT). (3) Population: we included studies examining the general human population, with no age limit. (4) Timing: no timeline settings. (5) Language: we included articles reported in English.

Finally, we reviewed 24 studies on CBCT/CT automatic landmark identification published before 2024 and summarised their characteristics (Table 1).

Patient selection

This study was approved by the Medical Ethics Committee of the First Affiliated Hospital of Fujian Medical University (approval no. MRCTA, ECFAH of FMU [2024]537). All the patients and their relatives provided written informed consent confirming their awareness that their imaging data would be used for clinical research. All procedures in the retrospective study were conducted in accordance with the approved guidelines, and all the participants' personal information was anonymised.

A total of 963 CBCT scans were screened. The inclusion criteria were as follows: (1) CBCT scans acquired with a voxel size of 0.25–0.4 mm^3 and a field of view (FOV) of at least 17×22 cm, and (2) inclusion of patients of all ages, sexes, and types of malocclusions and skeletal patterns. Exclusion criteria included: (1) imaging processes that exhibited significant scatter during restorative work and (2) patients with incomplete clinical data or who declined to participate. Finally, the sample comprised 498 CBCT volumes. All CBCT scans were anonymised and assigned new serial numbers.

All CBCT scans were obtained using an i-CAT next-generation machine (Imaging Sciences International, Hatfield, PA, USA) producing data with an FOV of 17×22 cm and a scan time of 26 s. The acquired datasets were stored in DICOM format with an isometric voxel size of 0.25–0.40 mm.

Manual annotation

Each CBCT dataset was imported into Dolphin 3D software (Dolphin Imaging and Management Systems, Chatsworth, CA, USA) and aligned to an anatomic frame of reference: (1) The horizontal axial plane (HP) was defined by the right porion, left porion, and the midpoint between the right and left orbitale; (2) The sagittal plane (SP) was perpendicular to the horizontal axial plane and included the nasion and sella; (3) The coronal plane (CP) was set perpendicular to the other two planes and passed through the sella (Fig. 1).

After reorientation, new DICOM files were generated for each dataset and imported into the Checkpoint software (ITK, Davis, Calif, USA) for manual landmark plotting and automatic detection using the proposed method. A total of 43 landmarks were used, comprising 5 cranial skeletal landmarks, 8 maxillary skeletal landmarks, 9 mandibular skeletal landmarks, 10 dental and dentoalveolar landmarks, and 11 soft tissue landmarks (Table 2 and Fig. 2).

To establish the ground truth (true position) for the selected landmarks, two judges independently performed manual landmark identification in the Checkpoint software, which had multiplanar reconstruction images in the axial, sagittal, and coronal views (Fig. 1). Judge 1 was an orthodontist, and Judge 2 was an oral and maxillofacial surgeon, both considered experts with over 10 years of experience in landmark identification in 2D and 3D imaging. In the first round of calibration, the two judges analysed 10 randomly selected CBCT images based on the operational definitions of the 43 selected landmarks. Following the first calibration, an additional 10 CBCT images were analysed by comparing the x, y, and z coordinates of the landmarks identified by both judges in a second round of calibration. After two rounds of calibration, the inter-judge reliability was found to exceed 0.99, and 43 landmarks were identified on the 498 CBCT images. The ground truth was established by calculating the mean values of the x, y, and z coordinates for each landmark across the identifications made by both judges.

Algorithm

To predict landmark positions in CBCT images, our model employed U-Net as its feature extraction framework and integrated a custom-designed Efficient Global Attention (EGA) module, which replaced the conventional high-dimensional feature extraction process. This modification improved the model's ability to capture global features more effectively. The model structure is shown in Fig. 3.

Given an input image $I \in R^{H \times W \times D}$, an additional channel dimension was first introduced, and a $1 \times 1 \times 1$ convolution layer (CL) was then applied to expand the number of channels. This process can be defined as follows:

$$X = CL(I)$$

where $X \in R^{H \times W \times D \times C}$ represents the output of CL.

The introduction of the CL facilitated subsequent calculations. This was followed by three stages: encoding, decoding, and prediction.

1) Encoding

Encoding consisted of two main processes. In the first process, we retained the traditional downsampling block

Table 1 Summary characteristics of studies on 3D-CT automatic landmark identification prior to 2024

Year, Author	Quantity of CBCT	Quantity of CT	The total number of landmarks (number of hard tissue landmarks)	Character of dentition	General method	Main Result	
						MRE ± SD(mm);	SDR < 2 mm
2014, Shahidi et al. [21]	28	\	14 (14)	Mixed and permanent dentiton	Atlas-based	3.40 mm	
2015, Gupta et al. [13]	30	\	20 (20)	Postgraduate orthodontic dentition	Knowledge-based	2.01 ± 1.23 mm	64.67%
2016, Zhang et al. [22]	41	30	15 (15)	No mention	Learning-based	1.44 mm	
2016, Gupta et al. [23]	30	\	21 (21)	No mention	Knowledge-based		
2017, Codari et al. [24]	18	\	21 (21)	Permanent dentition	Atlas-based		
2017, Zhang et al. [25]	77	30	15 (15)	No mention	Learning-based	1.10 ± 0.71 mm	
2018, Montufar et al. [26]	30	\	18 (18)	No mention	Learning-based	3.65 ± 1.43 mm	
2018, Montufar et al. [27]	24	\	18 (18)	No mention	Learning-based and Knowledge-based method	2.51 ± 1.61 mm	
2018, Neelapu et al. [28]	30	\	20 (20)	No mention	Knowledge-based	1.88 ± 1.10 mm	64.16%
2018, Jong et al. [29]	39	\	33 (33)	Permanent dentition	Learning-based		
2019, Lee et al. [30]	\	27	7 (7)	Permanent dentition	Learning-based	1.5 mm	90.5%
2019, O'Neil et al. [31]	\	20	22 (22)	No mention	Learning-based		
2019, Torosdagli et al. [32]	50	\	9 (9)	Permanent dentition	Learning-based		
2020, Ma et al. [33]	\	66	13 (13)	Permanent dentition	Learning-based	5.79 ± 0.98 mm	
2020, Zhang et al. [34]	77	30	15 (15)	No mention	Learning-based	1.10 ± 0.71 mm	
2022, Dot et al. [35]	\	198	33 (33)	Permanent dentition	Learning-based	1.0 ± 1.3 mm	90.4%
2022, Yun et al. [36]	\	24	90 (90)	Permanent dentition	Learning-based	2.88 mm	
2022, Ghowsi et al. [19]	100	\	53 (53)	Permanent dentition	Learning-based	3.19 ± 2.60 mm	35%
2022, Chen et al. [37]	89	33	17 (17)	No mention	Learning-based	1.64 ± 1.13 mm	74.28%
2022, Lang et al. [38]	50	45	105 (105)	Permanent dentition	Learning-based	1.38 ± 0.95 mm	
2023, Gillot et al. [20]	143	\	32 (32)	No mention	Learning-based	1.54 ± 0.87 mm	
2023, Xu et al. [39]	\	135	32 (32)	Mixed and permanent dentiton	Learning-based	1.46 ± 1.31 mm	90%
2023, Xu et al. [40]	\	117	27 (27)	Mixed and permanent dentiton	Learning-based	1.33 mm	
2023, Tao et al. [15]			77 (64)	Permanent dentition	Learning-based	1.81 ± 0.89 mm	76.62%

(DB) to extract local features. The DB in our model was defined as follows:

$$X = Conv(MaxPool(X))$$

where $MaxPool(\cdot)$ represents the max-pooling layer, and $Conv(\cdot)$ denotes two 3D CLs. Each CL employed a $3\times3\times3$ kernel, stride of 2, and padding of 1, reducing spatial dimensions while preserving local features. In the

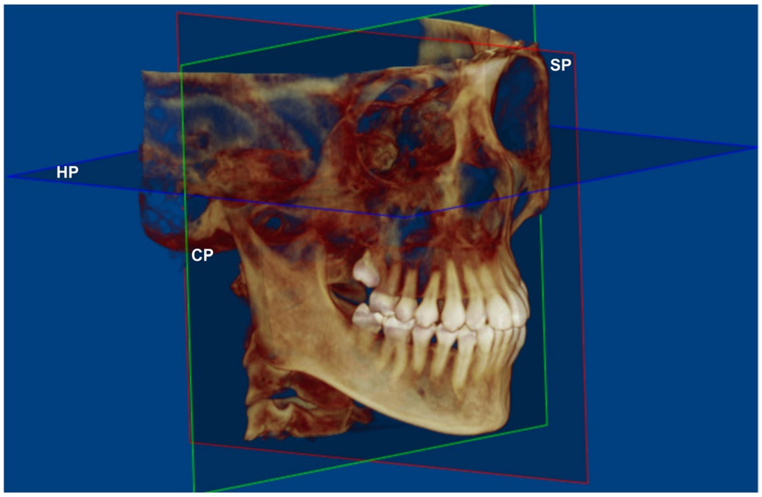

Fig. 1 Anatomic frame of reference in cone-beam computed tomography (CBCT). The horizontal axial plane (HP) was defined by the right porion, left porion, and the midpoint between the right and left orbitale. The sagittal plane (SP) was perpendicular to the horizontal axial plane and included the nasion and sella. The coronal plane (CP) was set perpendicular to the other two planes and passed through the sella

second process, we employed our custom-designed EGA module to effectively capture global information. Figure 4 illustrates its structure in detail. The core computational process in this module can be defined as follows:

$$X = EGA(X)$$

The EGA module incorporates three essential functions: dynamic position encoding [41], multi-scale global attention, and feature fusion network [42]. This module increases the diversity of generated global features, thereby enhancing the model's sensitivity to landmark locations.

2) Decoding

During decoding, feature maps from each encoder stage were first reused. The spatial sizes of the fused feature were progressively expanded, while the number of channels was gradually reduced, ultimately aligning with the dimensions of the original input features. The upsampling block in our model can be defined as follows:

$$X = Conv(Concat(\tau(X_d), X_e, dim = c))$$

where X_d and X_e represent the decoder and encoder features, respectively.$\tau(\cdot)$ represents trilinear interpolation, which doubles the resolution. X_d and X_e were concatenated along the channel dimension and passed through a $1\times1\times1$ CL with a stride of 1 to complete the upsampling process.

3) Prediction

In the prediction phase, through the encoder and decoder feature extraction modules, we obtained feature maps containing both local and global information at the same resolution as that of the original input images. A $1\times1\times1$ convolutional prediction head was then employed to convert these feature maps into N heat maps, each corresponding to one of the N landmark points. For each heat map, the location of the maximum value was identified, and its coordinates were considered the final predicted landmark position.

Evaluation of accuracy of the proposed algorithm

To evaluate the overall localisation performance on our test set, we calculated two common criteria for each landmark [43]: 1) mean radial error (MRE), which is the mean Euclidean distance between the reference and the predicted landmark; and 2) success detection rate (SDR), which refers to the proportion of landmarks with radial errors below 2 mm, 3 mm, and 4 mm. These metrics are widely used in landmark detection tasks, and their formulas are as follows:

$$\mathrm{MRE} = \frac{\sum_{k=1}^{n}\sqrt{(x_k - \widehat{x}_k)^2 + (y_k - \widehat{y}_k)^2 + (z_k - \widehat{z}_k)^2}}{n}$$

where x_k, y_k, and z_k represent the manually annotated landmark coordinates along the x, y, and z axes, respectively, and $\widehat{x}_k, \widehat{y}_k$, and $\widehat{z}_k$ denote the estimated landmark coordinates along the x, y, and z axes, respectively.

$$\mathrm{SDR} = \frac{\sum_{k=1}^{N_{Total}} 1(d_k < \sigma)}{N_{Total}} \times 100\%$$

where d_k represents the localisation error for the k th landmark, typically calculated as the Euclidean distance

Table 2 Definition of 3D landmarks

No	Anatomic Region	Median/Bilateral	3D Notation	Landmark	Definition
1	Cranial base	Median	S	Sella	Midpoint of the fossa for the pituitary gland
2	Cranial base	Median	Na	Nasion	Midline structure that is the most anterior point of the frontonasal suture
3	Cranial base	Median	Ba	Basion	Midline structure found on the anterior region of the foramen magnum
4	Maxilla	Median	A	Point A	Midline point that is the deepest concavity on the maxilla between the ANS and prosthion
5	Maxilla	Median	Spr	Superior prosthion	Median point at the inferior tip of the septum between the maxillary central incisors
6	Mandible	Median	B	Point B	Midline structure that is the deepest concavity on the mandibular symphysis between the infradentale and pogonion
7	Mandible	Median	Id	Infradentale	Median point at the superior tip of the septum between the mandibular central incisors
8	Mandible	Median	Pog	Pogonion	Midline strucutre that is the most anterior part of the bony chin
9	Mandible	Median	Gn	Gnathion	Median point halfway between Po and Me; bisecting two lines, one following vertical margin of ramus and one following horizontal margin of corpus of mandible
10	Mandible	Median	Me	Menton	Midline strucutre that is the most inferior part of the bony chin
11	Maxilla	Median	ANS	Anterior nasal spine	Midline structure that is the most anterior point of the anterior nasal spine
12	Maxilla	Median	PNS	Posterior nasal spine	Midline structure that is the most posterior point of the posterior nasal spine
13	Teeth	Median	Uli	Upper 1 tip or incisal edge	Maxillary incisal tip (U1); Upper central incisor's most inferior point on the incisal edge
14	Teeth	Median	Ula	Upper 1 root apex	Upper central incisor's root apex
15	Teeth	Median	Lli	Lower 1 tip or incisal edge	Mandibular Incisor Tip (L1); Lower central incisor's most superior point on the incisal edge
16	Teeth	Median	Lla	Lower 1 root apex	Lower central incisor's root apex
17&18	Maxilla	Bilateral	Or	Orbitale	Most inferior point on the infraorbital rim
17		Bilateral	Or-R		
18		Bilateral	Or-L		
19&20	Temporal bone	Bilateral	Po	Porion	Most superior point on the upper margin of the external auditory meatus
19		Bilateral	Po-R		
20		Bilateral	Po-L		
21&23	Mandible	Bilateral	Co	Condylion	The most superior point of the mandibular condyle
21		Bilateral	Co-R		
23		Bilateral	Co-L		
22&24	Mandible	Bilateral	Go	Gonion	Point where the ramus and mandibular planes intersect and the most posterior, inferior, and lateral point at the angle of the mandible
22		Bilateral	Go-R		
24		Bilateral	Go-L		
25&28	Teeth	Bilateral	U6	Upper 6	Occlusal contact points of upper 6
25		Bilateral	U6-R		
28		Bilateral	U6-L		
26&29	Teeth	Bilateral	U6-B	Upper 6-root bifurcation	Upper right 6-root bifurcation and upper left 6-root bifurcation
26		Bilateral	UR6-B		
29		Bilateral	UL6-B		
27&30	Teeth	Bilateral	L6-B	Lower 6-root bifurcation	Lower right 6-root bifurcation and lower left 6-root bifurcation

Table 2 (continued)

No	Anatomic Region	Median/Bilateral	3D Notation	Landmark	Definition
27		Bilateral	LR6-B		
30		Bilateral	LL6-B		
31&32	Maxilla	Bilateral	Py	Pyriform	Anterolateral margin of the nasal aperture represented by the most anteromedial point on the maxilla, forming the bony alar base
31		Bilateral	Py-R		
32		Bilateral	Py-L		
33&34	Soft Tissue	Bilateral	Alare	Alare	The most lateral point on the base of nasal ala
33		Bilateral	Alare-R		
34		Bilateral	Alare-L		
35	Soft Tissue	Median	Ns	Nasion of soft tissue	Point directly anterior to the nasofrontal suture, in the mid-line,
36	Soft Tissue	Median	Prn	Pronasale	The most anteriorly protruded point of the apex nose; in the case of a bifid nose, the more protruding tip is cho-sen
37	Soft Tissue	Median	Sn	Subnasale	Median point at the junction between the lower border of the nasal septum and the philtrum area
38	Soft Tissue	Median	UL’	Labiale superius	Midpoint of the vermilion border of the upper lip
39	Soft Tissue	Median	Sto	Stomion	Midline point of the labial fissure when the lips are closed naturally with teeth shut in the natural position
40	Soft Tissue	Median	LL’	Labiale inferius	Midpoint of the vermilion border of the lower lip
41	Soft Tissue	Median	Pogs	Pogonian of soft tissue	Most anterior midpoint of the chin, located on the skin
42	Soft Tissue	Median	Gns	Gnathion of soft tissue	Median point halfway between Pogs and Mes
43	Soft Tissue	Median	Mes	Menton of soft tissue	Most inferior median point of the chin, located on the skin

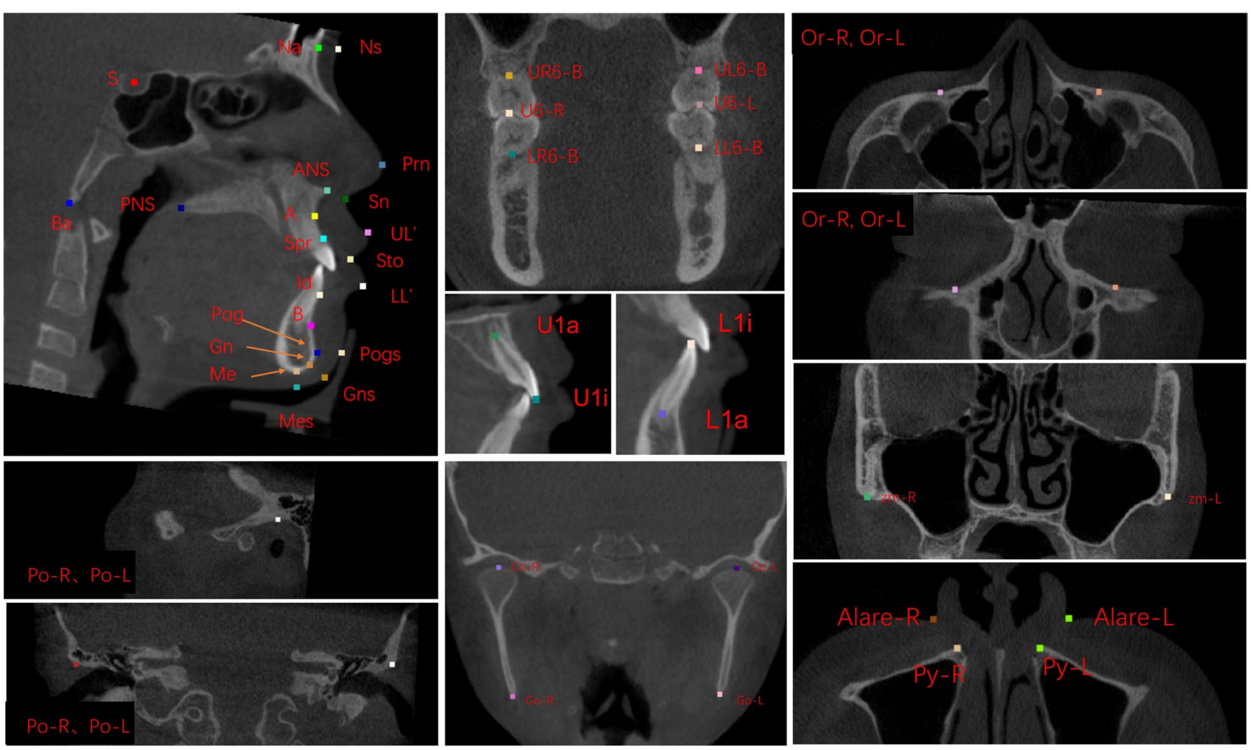

Fig. 2 Landmarks in axial, sagittal, and coronal slices of cone-beam computed tomography (CBCT)

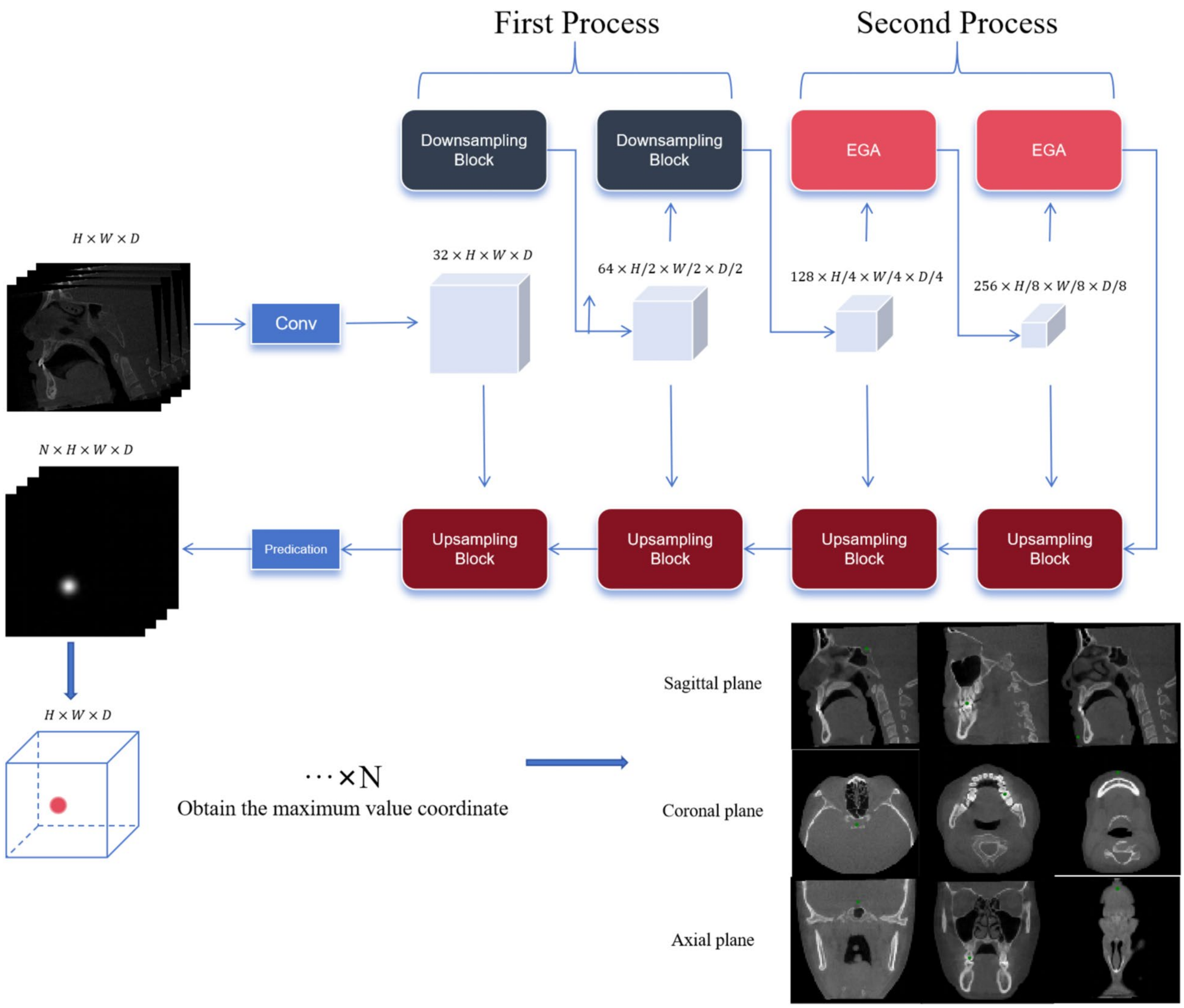

Fig. 3 Structure of the proposed model

between the predicted and ground truth landmark positions. The indicator function $1(d_k < \sigma)$ equals 1 if $d_k < \sigma$ and 0 otherwise, meaning that the k th landmark is considered successfully detected if the error falls below the threshold σ.

Statistical analysis

The interobserver intraclass correlation coefficient (ICC) for each landmark was calculated from two sets of manual observations using the Statistical Package for the Social Sciences (SPSS, Chicago, Ill, USA). The Euclidean distances between manually and automatically identified coordinates were calculated and compared.

Results

Composition of the dataset

We retrospectively collected approximately 963 CBCTs from the First Affiliated Hospital of Fujian Medical University over the past 7 years. These data were screened based on inclusion and exclusion criteria, resulting in 498 eligible cases of the 498 patients, 246 were male and 252 were female, with ages ranging from 4 to 46 years and a mean age of 17.64 ± 5.44 years. Categorisation by dentition revealed that eight patients had deciduous dentition, 238 had mixed dentition, and 255 had permanent dentition. According to the measurement and analysis of the matched lateral radiographs, 136 cases were classified as skeletal class I, 218 as class II, and 144 as class III (Table 3). The dataset also included 23 patients with

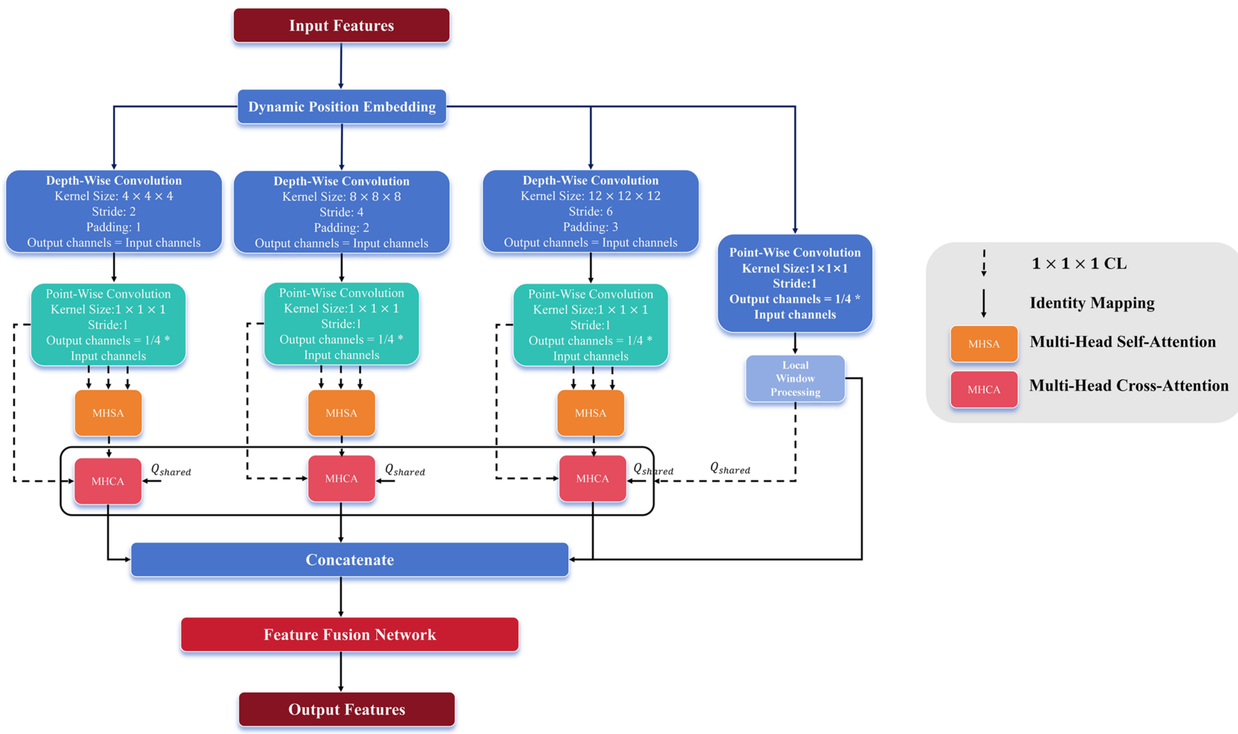

Fig. 4 Structure of the Efficient Global Attention (EGA) module

Table 3 Composition of the dataset

Classification	Deciduous dentition (8)	Mixed dentition (238)	Permanent dentition (252)	Total (498)
Skeletal class I	1	49	86	136
Skeletal class II	4	112	102	218
Skeletal class III	3	77	64	144

cleft lip and palate, 38 with condylar fractures, and 2 with Crouzon syndrome. In 42 cases, high lip muscle tension prevented contact between the upper and lower lips, and 65 cases of permanent dentition exhibited varying degrees of dentition defects. Among the eight primary dentition CBCT cases, two were of Crouzon syndrome, and six involved condylar fractures. All CBCT scans were conducted according to the ALARA principle.

Experimental settings

The datasets were divided into training and test sets at a ratio of 5:1. Each image contained 43 landmarks and was saved in the nii.gz format. The low-resolution images had a minimum size of $350 \times 350 \times 350$ with a voxel spacing of 0.4 mm, while the high-resolution images had a minimum size of $650 \times 650 \times 650$ with a voxel spacing of 0.25 mm. All images were resampled to a uniform resolution of $96 \times 96 \times 96$ using trilinear interpolation, and the landmark coordinates were proportionally scaled according to the respective image resolutions. The model's hyperparameters were set as follows: batch size of 6, Gaussian kernel size for the heat maps of 21, sigma of 2, and a total of 200 training epochs. The AdamW optimiser was used with the default settings.

The proposed models were trained on a server equipped with an Intel Xeon Gold 6325 central processing unit (2.90 GHz) and an NVIDIA A40 graphics processing unit (GPU) with 48 GB of memory, using cuDNN version 8.9.0.2 for GPU acceleration. The implementation was carried out using the PyTorch framework, version 2.2.2. Each landmark prediction took approximately 4.2 s on the GPU. The prediction of the test scans required 8.8 GB of cache memory and 2.1 GB of GPU memory. Each agent performed an average of 90 moves to reach the landmark position.

Results of reference landmark detection

The interobserver ICC for each landmark was excellent (> 0.9). The 3D coordinates of landmarks detected automatically were compared with those identified manually for each of the 498 image datasets. The Euclidean distances between the coordinates obtained via manual identification and automatic detection using the proposed algorithm were calculated.

Table 4 Mean absolute error at each coordinate, MRE with SD of each landmark, and SDRs for 2-, 3-, and 4-mm range criteria

No	Landmark	SDR < 2.0mm	SDR < 3.0mm	SDR < 4.0mm	MRE Mean (mm)	MRE SD (mm)	x-axis Mean (mm)	x-axis SD (mm)	y-axis Mean (mm)	y-axis SD (mm)	z-axis Mean (mm)	z-axis SD (mm)
1	S	65.00%	93.33%	100.00%	1.69	1.01	0.67	0.93	0.62	0.89	0.80	0.88
2	Na	55.56%	83.33%	94.44%	1.95	1.16	0.59	0.84	0.46	0.80	1.35	1.21
3	Ba	56.67%	91.67%	100.00%	1.79	1.05	1.14	1.01	0.63	0.87	0.54	0.74
4	A	71.67%	96.67%	98.33%	1.36	1.03	0.35	0.70	0.46	0.80	0.80	0.91
5	Spr	73.33%	96.67%	100.00%	1.58	0.85	0.55	0.81	0.51	0.83	0.79	0.84
6	B	60.00%	83.33%	100.00%	1.91	1.01	0.72	0.86	0.49	0.83	1.17	1.07
7	Id	65.00%	95.00%	98.33%	1.40	1.18	0.72	0.86	0.43	0.79	0.68	0.90
8	Pog	58.33%	95.00%	100.00%	1.69	0.92	0.79	0.88	0.49	0.83	0.88	0.78
9	Gn	71.67%	95.00%	98.33%	1.33	1.11	0.73	0.93	0.43	0.78	0.52	0.75
10	Me	71.67%	96.67%	100.00%	1.42	1.07	0.78	0.98	0.58	0.86	0.33	0.63
11	ANS	73.33%	95.00%	98.33%	1.33	1.08	0.35	0.70	0.80	0.99	0.46	0.72
12	PNS	55.00%	93.33%	100.00%	1.83	0.94	0.72	0.86	0.96	0.94	0.62	0.91
13	Uli	75.00%	95.00%	98.33%	1.38	1.19	0.76	0.88	0.46	0.81	0.50	0.93
14	Ula	55.00%	86.67%	93.33%	2.00	1.16	0.67	0.86	0.95	1.10	0.96	1.06
15	Lli	81.67%	98.33%	100.00%	1.24	0.99	0.58	0.83	0.37	0.74	0.51	0.74
16	Lla	60.00%	93.33%	100.00%	1.68	0.95	0.58	0.83	0.68	0.91	0.76	0.93
17	Or-R	40.00%	75.00%	93.33%	2.28	1.51	1.72	1.51	0.67	0.89	0.65	0.78
18	Or-L	58.33%	86.67%	96.67%	2.01	1.23	1.58	1.16	0.53	0.94	0.34	0.66
19	Po-R	51.67%	86.67%	95.00%	1.97	1.18	0.83	1.02	0.91	1.06	0.84	0.95
20	Po-L	43.33%	83.33%	93.33%	2.22	1.14	0.97	0.99	0.96	1.17	1.05	0.97
21	Co-R	38.33%	70.00%	88.33%	2.40	1.28	0.73	0.87	1.64	1.34	0.85	0.96
22	Go-R	50.00%	93.33%	86.67%	2.26	1.53	0.61	0.84	1.33	1.41	1.15	1.15
23	Co-L	63.33%	73.33%	98.33%	1.66	1.30	0.93	1.13	0.61	0.94	0.53	0.88
24	Go-L	45.00%	93.33%	96.67%	2.05	1.40	0.58	0.89	1.14	1.09	1.13	1.13
25	U6-R	68.33%	90.00%	91.67%	1.57	1.25	0.52	0.86	0.75	1.16	0.69	0.80
26	UR6-B	65.00%	95.00%	98.33%	1.61	0.96	0.50	0.80	0.43	0.87	0.97	0.89
27	LR6-B	75.00%	98.33%	100.00%	1.25	1.06	0.36	0.72	0.62	0.88	0.55	0.77
28	U6-L	69.44%	91.67%	100.00%	1.57	1.06	0.58	0.84	0.70	1.00	0.57	0.87
29	UL6-B	53.33%	86.67%	98.33%	1.72	1.26	0.59	0.84	0.70	0.90	0.98	1.11
30	LL6-B	73.33%	98.33%	98.33%	1.37	1.17	0.50	0.92	0.49	0.82	0.74	0.82
31	Py-R	50.00%	90.00%	98.33%	1.87	1.20	1.00	1.10	0.39	0.76	1.07	0.94
32	Py-L	53.33%	100.00%	100.00%	2.01	1.10	1.03	0.99	0.46	0.81	1.12	1.06
33	Alare-R	56.67%	100.00%	100.00%	1.80	1.01	0.57	0.84	0.64	0.88	1.03	1.01
34	Alare-L	60.00%	100.00%	100.00%	1.79	1.08	0.70	0.86	0.49	0.90	1.09	0.96

Table 4 (continued)

No	Landmark	SDR < 2.0mm	SDR < 3.0mm	SDR < 4.0mm	MRE Mean (mm)	MRE SD (mm)	x-axis Mean (mm)	x-axis SD (mm)	y-axis Mean (mm)	y-axis SD (mm)	z-axis Mean (mm)	z-axis SD (mm)
35	Ns	58.33%	100.00%	96.67%	1.93	1.36	0.45	0.85	0.55	0.85	1.33	1.36
36	Prn	76.67%	98.33%	100.00%	1.30	0.95	0.49	0.79	0.40	0.76	0.65	0.78
37	Sn	73.33%	100.00%	100.00%	1.37	0.93	0.46	0.78	0.36	0.74	0.80	0.79
38	UL’	56.67%	83.33%	100.00%	1.68	1.18	0.46	0.77	0.36	0.73	1.26	1.08
39	Sto	73.33%	100.00%	98.33%	1.29	1.01	0.49	0.78	0.58	0.86	0.49	0.73
40	LL’	60.00%	100.00%	95.00%	1.82	0.98	0.66	0.85	0.65	0.90	1.01	0.95
41	Pogs	31.67%	88.33%	93.33%	2.86	1.46	0.87	0.93	1.26	1.13	1.76	1.67
42	Gns	40.00%	75.00%	100.00%	2.39	1.12	0.85	0.99	1.25	0.98	1.25	1.09
43	Mes	53.33%	70.00%	100.00%	2.03	1.02	0.73	0.93	0.71	1.00	1.05	1.09
Average	Hard tissue	60.83%	90.63%	97.27%	1.73	1.14	0.74	0.91	0.68	0.93	0.78	0.90
	Soft tissue	58.18%	92.27%	98.48%	1.84	1.10	0.61	0.85	0.66	0.89	1.07	1.05
	All	60.16%	91.05%	97.58%	1.76	1.13	0.71	0.90	0.67	0.92	0.85	0.94

MRE Mean Radial Error, *SDR* Successful Detection Rate

Table 4 presents the mean absolute error in the x, y, and z axes; MRE; and SDR within 2-, 3-, and 4-mm error ranges of manual marking for each of the 43 detected landmarks. The overall mean absolute error for the 43 landmarks was 0.71 mm in the x axis, 0.67 mm in the y axis, and 0.85 mm in the z axis. The overall MRE for the 43 landmarks was 1.76 ± 1.13 mm, and the SDR was 60.16%, 91.05%, and 97.58% within 2-, 3-, and 4-mm error ranges of manual marking, respectively. Of the 43 landmarks, 32 were hard tissue markers, and 11 were soft tissue markers. The average MRE for hard tissue landmarks was 1.73 mm, and that for soft tissue landmarks was 1.84 mm, all of above meeting the criteria for clinical accuracy (a distance of 2 mm or less between automated and manual recognition is considered clinically accurate) [13]. Fifteen hard tissue landmarks (25/32, 78.1%) and seven soft tissue landmarks (8/11, 72.7%) had an MRE of less than 2 mm.

L1i was detected with a minimum MRE of 1.24 mm, and Pogs was detected with a maximum MRE of 2.86 mm among the 43 landmarks.

Discussion

This study presents an innovative approach for achieving robust and precise localisation of anatomical landmarks on CBCT for individuals of different ages and with various malocclusion types. The mean MRE of soft tissue, hard tissue, and the total 43 landmarks were 1.76 mm, 1.73 mm, and 1.84 mm, respectively, meeting the criteria for clinical accuracy.

The diagnosis of dento-maxillofacial deformities should be made in 3D. In conformity with the principle and under the condition when the benefits of the diagnosis and/or treatment plan outweigh the potential risks of exposure to radiation [7–11], CBCT offers a more comprehensive multi-angle view, higher spatial resolution, and finer anatomical details than 2D lateral films, contributing to a more accurate diagnosis, evaluation, and surgical planning for malocclusion. However, manual identification of anatomical landmarks in 3D-CT is challenging, time-consuming, and dependent on the clinician's expertise. However, our proposed algorithm required only 4.2 s on the GPU for each landmark prediction and the precision has reached the standard of clinical accuracy, hopefully improving access to accurate diagnosis in underserved areas or reducing healthcare costs by supported online remote detection.

In recent years, various methods have been proposed for automatic 3D-CT landmark identification; learning-based methods have demonstrated higher accuracy and currently dominate research in this area [12, 14]. The datasets used in most previous learning-based automatic craniomaxillofacial landmarks detection studies, although multicentre [18, 20], predominantly focus on certain patient types respectively: patients with permanent dentition and individual normal occlusion and without craniofacial anomalies or syndromes [19, 20, 28–30], patients with cleft lip and palate [40], patients with hemifacial microsomia [39], and patients requiring orthognathic surgery for dentomaxillofacial deformities [15, 18, 26, 32, 34, 35, 44]. To better simulate routine clinical practice and enhance the robustness, we incorporated the CBCT data, including various dentition phases, malocclusion types, and anatomical structural deficiencies (such as impacted teeth, fractured condyles, and cleft lip and palate) in our study (Table 3).

Accurate and reproducible identification of anatomical landmarks is not only a crucial step in assessing and quantifying anatomical abnormalities [45] but also forms the foundation for AI and deep learning applications [46]. The irregular and varied nature of the dataset, both anatomically and morphologically, likely enhanced the algorithm's robustness while also impacting the overall recognition accuracy. In our paper, the 2-mm SDR and MRE of Co-L were 63.33% and 1.66 mm, respectively, while those for Co-R were only 38.33% and 2.40 mm, respectively. This disparity may be attributed to the inclusion of CBCT scans from 38 patients with condyle fractures, 22 of whom had right condyle fractures, 5 had bilateral fractures, and only 11 had left fractures. Consequently, the difficulty in identifying Co post-fracture likely contributed to this result. The irregular anatomy of the abnormal side resulted in a decrease in the automatic identification accuracy when compared to that of the normal side. This finding is corroborated by the study conducted by Xu et al. [39]. Unlike Shahidi et al. [21], who used 28 CBCTs from individuals aged 10 years and older, and Dot et al. [35], who included 198 permanent dentitions in their study, reporting similar accuracy for apical and incisal points in the upper and lower incisors, our study found the lowest SDR and highest MRE of all incisor-related anatomical points at point U1a (MRE: 2.00 mm) compared to the average of U1i + L1i + L1a (MRE: 1.43 mm). This discrepancy may stem from the fact that nearly half of our patients had deciduous and mixed dentition (246/498, 49.4%), with most maxillary apical foramina incompletely closed, forming trumpet-shaped openings of varying sizes, making it challenging to accurately locate U1a, this is similar to the findings of Park et al. [47].

To better identify different types of craniofacial structures from diversity datasets, thus improving the accuracy and generalisation of the algorithm, our proposed algorithm utilised a UNet architecture as the backbone, allowing for feature extraction at higher resolutions and minimising information loss during image downsampling. This preserves more detailed features. In comparison to the traditional UNet model, we integrated an EGA module during high-dimensional feature extraction to effectively capture global information and enhance the model's feature representation capability. As a result, although containing a large number of irregular condylar shapes, our mean MRE of Co was 2.02 mm, which was more accurate than previous literature where the AI predicted position of Co was 2.79 mm and 3.56 mm from the correct position [13, 28]. Thus, improving the accuracy of detecting anatomical irregularities significantly improved the overall accuracy in our study.

Additionally, within the hard tissue landmarks, Po and Or were less accurately identified in this study. Due to the anatomical morphology, the absence of a distinct curvature makes manual identification less reproducible, resulting in a reduction in SDR [12]. The FH plane, formed by Po and Or, is commonly used for assessing the type and severity of maxillofacial deformities and is one of the most commonly utilised reference planes. However, landmark reproducibility does not necessarily equate to plane reproducibility, as the latter depends on the direction of landmark errors [48]. In our study, the bias of these four points was primarily along the x axis. The mean error on the z axis was 0.72 mm. Therefore, the accuracy remains clinically suitable, and these results are consistent with those of other studies [19, 49].

Due to the heterogeneity of datasets across studies in terms of types, sources, and inclusion criteria, as well as variability in the number and types of labelled points, direct comparison of the accuracy of different algorithms based on MRE or SDR values is not feasible. Notably, the Na point, which shows a high 2-mm SDR in 2D lateral cephalometry [4], exhibited a lower SDR in our study (55.56% SDR for 2 mm). Despite its anatomical definition allowing for straightforward AI probing [50], the MRE for Na in this study (1.95 mm) was higher than the average MRE (1.76 mm). After reviewing literature from the past 5 years on automatic 3D-CT landmarking (learning-based approaches) using MRE as an evaluation metric, we found that the MRE for Na was consistently higher than the average when CBCT was used [19, 37], a finding aligning with our results. However, when CT datasets were used, the MRE of Na was below the average of all points [30, 35, 36, 39, 40]. This could be due to the restricted FOV in CBCT, potentially placing Na at the border of or outside the image data, thus impacting the precision of automated identification. In contrast, CT provides a more comprehensive view, eliminating ambiguity in Na position and enhancing accuracy. Interestingly, studies published before 2018 that relied solely on CBCT as the data source reported that the MRE of Na was less than the average [13, 21, 26–28]. This may be due to the limited number of CBCT datasets (28–41), which all adhered to established anatomical and morphological rules. These findings support our hypothesis that the ability of learning algorithms to discern anatomical landmarks and determine precise positioning depends on the availability of high-quality datasets. We propose a bold direction for future research: could using CT images with complete and well-defined anatomical landmarks as the primary training set improve the precision and accuracy of algorithms when validating CBCT images of varying specifications and sizes? We plan to explore this hypothesis in future studies.

There are two methods for 3D-CT calibration. To enhance the accuracy of landmark identification, we employed a multi-view angle MRP calibration method, which has demonstrated higher clinical accuracy; however, this method requires more than twice the time by human experts when compared to direct calibration on the rendered models [51] (Fig. 2). In contrast to the majority of previous studies that identified landmarks on reconstructed 3D skull models [12, 14, 16, 17], our algorithm directly recognises them on the MRP layer, obviating the need for secondary reconstruction of imaging data. The methodology in this study enables more precise localisation of key points within different hard tissues, such as root bifurcation points, which are a pivotal element in the Yonsei CBCT transverse analysis method and serve as a crucial diagnostic indicator of maxillary transverse discrepancy [5, 52]. The mean MRE at the first molar root bifurcation point in this study was 1.49 mm. This accuracy meets the standards for clinical relevance and offers promise in addressing the limitations of previous 2D and many 3D automatic identification techniques, which are confined to vertical and sagittal diagnoses. Consequently, the clinical application of the proposed algorithm can be further expanded.

To our knowledge, previous research on 3D-CT automatic landmark localisation has mainly focused on hard tissues [12, 14, 16–18] (Table 1), while a few articles on soft tissue identification were basically limited to patients with orthognathic surgery-related maxillofacial anomalies [15, 44, 53]. This may be because some algorithmic models consider soft tissue images as noise, thereby reducing algorithmic accuracy and leading to their frequent exclusion [33]. Moreover, although CBCT has a lower radiation dose, its soft-tissue resolution is inferior to that of CT. With the continuous improvement in CBCT accuracy, some researchers have recently started using CBCT for facial soft tissue research [54, 55]. In this study, we aimed to obtain as much information as possible from limited data; therefore, we attempted automatic soft-tissue detection to patients of all ages in CBCT. Although the accuracy of landmarks such as Na was affected by the boundaries of CBCT images. A similar issue arose in the soft tissue region of the chin, where the accuracy of Pogs, Gns, and Mes points was compromised by the chin rest used to hold the head in position. Nevertheless, the average MRE of the soft tissue points in our study was 1.84 mm, and the SDR values were 58.18%, 92.27%, and 98.48% within 2, 3, and 4 mm, respectively. These findings suggest great potential for automated soft tissue landmark detection with CBCT, with these data serving as valuable adjunct to clinical examination.

This study has certain limitations. In this study, we used only the proposed algorithm to train and test the dataset. To further verify its effectiveness, we plan to test the same dataset with other related algorithms in subsequent experiments for a more intuitive comparative analysis. Thus, additional heterogeneous multi-centre CBCT dataset should be used to further verify the generalisation of AI network.

Conclusion

The mean error distance between automated and manual recognition in our study was 1.76 ± 1.13 mm, resulting in an SDR of 97.58% within a 4- mm error range, which is considered a clinically accurate level for conventional cephalometric measurements. This algorithm demonstrated good robustness and clinically acceptable precision in automatic detecting soft and hard tissue landmarks for all age groups and various types of skeletal malocclusions. In addition to the landmarks that are conventionally required for traditional sagittal and vertical diagnostics in cephalometrics, the automatic recognition of anatomical markers related to the transverse discrepancy of jaw width was realised. This study lays the foundation for achieving fully automated diagnosis of 3D craniomaxillofacial malformations.

Abbreviations

CBCT	Cone-Beam Computed Tomography
3D	Three-Dimensional
MRE	Mean Radial Error
SDR	Successful Detection Rate
AI	Artificial Intelligence
2D	Two-Dimensional
FOV	Field of View
DICOM	Digital Imaging and Communications in Medicine
CL	Convolution Layer
DB	Downsampling Block
EGA	Efficient Global Attention
ICC	Intraclass Correlation Coefficient
SPSS	Statistical Package for the Social Sciences
GPU	Graphics Processing Unit

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Authors' contributions
Yan Jiang, Shuli Xing, and Lisong Lin contributed to the study conception and design. Yan Jiang, Canyang Jiang, You Wu, Hao Liang and Li Huang performed data collection and analysis. Yan Jiang and Canyang Jiang wrote the first draft of the manuscript, and all authors commented on the previous versions. All the authors have read and approved the final version of the manuscript.

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Data availability
The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate
This study was approved by the Medical Ethics Committee of the First Affiliated Hospital of Fujian Medical University (approval no. MRCTA, ECFAH of FMU [2024]537). Our study adhered to the Declaration of Helsinki. All patients and their relatives provided written informed consent before inclusion in this study.

Consent for publication
Not Applicable.

Competing interests
The authors declare no competing interests.

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